

Dietary nitrate reduces maximal oxygen consumption while maintaining work performance in maximal exercise.

The anion nitrate-abundant in our diet-has recently emerged as a major pool of nitric oxide (NO) synthase-independent NO production. Nitrate is reduced stepwise in vivo to nitrite and then NO and possibly other bioactive nitrogen oxides. This reductive pathway is enhanced during low oxygen tension and acidosis. A recent study shows a reduction in oxygen consumption during submaximal exercise attributable to dietary nitrate. We went on to study the effects of dietary nitrate on various physiological and biochemical parameters during maximal exercise. Nine healthy, nonsmoking volunteers (age 30+/-2.3 years, VO(2max) 3.72+/-0.33 L/min) participated in this study, which had a randomized, double-blind crossover design. Subjects received dietary supplementation with sodium nitrate (0.1 mmol/kg/day) or placebo (NaCl) for 2 days before the test. This dose corresponds to the amount found in 100-300 g of a nitrate-rich vegetable such as spinach or beetroot. The maximal exercise tests consisted of an incremental exercise to exhaustion with combined arm and leg cranking on two separate ergometers. Dietary nitrate reduced VO(2max) from 3.72+/-0.33 to 3.62+/-0.31 L/min, $P < 0.05$. Despite the reduction in VO(2max) the time to exhaustion trended to an increase after nitrate supplementation (524+/-31 vs 563+/-30 s, $P = 0.13$). There was a correlation between the change in time to exhaustion and the change in VO(2max) ($R(2) = 0.47$, $P = 0.04$). A moderate dietary dose of nitrate significantly reduces VO(2max) during maximal exercise using a large active muscle mass. This reduction occurred with a trend toward increased time to exhaustion implying that two separate mechanisms are involved: one that reduces VO(2max) and another that improves the energetic function of the working muscles. Copyright 2009 Elsevier Inc. All rights reserved.